Computer Organization and Architecture
Assignment 1

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Submitted to:
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Q1. What is Raspberry Pi ?

Answer:
Raspberry Pi is a small and low-cost single-board computer. It is used to learn programming, build electronics projects, and develop robotics and IoT applications. It can also connect with sensors and other electronic devices through its GPIO pins.

Q2. What are the 10 takeaways of the AI Summit?

Answer:

- AI is moving from research to real-life applications.
- Building safe and responsible AI is becoming more important.
- Many countries are investing in AI to support future growth.
- Better computing infrastructure is needed for AI development.
- AI skills will be valuable for future careers.
- AI can improve productivity and reduce repetitive work.
- Governments, companies, and researchers should work together on AI.

- Privacy, security, and transparency should be considered while developing AI.
- Continuous research and innovation are needed to make AI better.
- AI will have a major impact on technology, business, and society.

Q3. What is YOTTA?

Answer:
YOTTA is an Indian technology company that provides cloud computing, AI services, and data center solutions. It helps businesses store data, run applications, and manage their digital infrastructure securely.

Q4. What is NPTEL?

Answer:
NPTEL (National Programme on Technology Enhanced Learning) is an online learning platform started by the IITs and IISc. It offers free courses, video lectures, assignments, and certification in many subjects.

Q5. How do you choose a programming language to design an application, and what are the important parameters?

Answer:
To choose a programming language, you should consider a few important things:

- What type of application you want to develop.
- How much speed and performance the application needs.
- Whether the language is secure and reliable.
- Which operating systems or platforms it should support.
- How easy it is to write and maintain the code.
- Whether useful libraries and tools are available.

- If there is good community support for solving problems.
- The cost of development and future maintenance.
  For example, C is used for system programming, Java is used for desktop and Android applications, and Python is used for AI and data science. Choosing the right language helps develop the application more efficiently.

Q6. What is the meaning of RAM size and word size? If the system is 64-bit, what is the speed and operating system of your PC?

Answer:

RAM Size: RAM (Random Access Memory) is the temporary memory used by a computer to store data while programs are running. More RAM allows the system to run multiple applications smoothly.

Word Size: Word size is the amount of data a processor can handle in one operation. A 64-bit processor can process 64 bits of data at a time and supports larger amounts of memory.

My PC Specifications:

- RAM Size: 16 GB
- Word Size: 64-bit (x64-based processor)
- CPU Speed: 2.00 GHz (Base Speed)
- Operating System: Windows 11 Home Single Language

Q7. Why is the GPU faster than the CPU?

Answer:
A GPU is faster than a CPU because it has many small cores that can perform many calculations at the same time. A CPU has fewer but more powerful cores and

usually performs tasks one by one. This makes the GPU better for graphics, gaming, and AI-related work.

Q8. Define CPU, GPU, and TPU.

Answer:

CPU (Central Processing Unit)	GPU (Graphics Processing Unit)	TPU (Tensor Processing Unit)
Used for general-purpose computing tasks.	Used for graphics processing and parallel computing.	Used mainly for artificial intelligence and machine learning.
Has a few powerful cores.	Has many smaller cores for handling multiple tasks at once.	Has specialized hardware designed for AI calculations.
Runs the operating system and applications.	Commonly used in gaming, video editing, and AI.	Commonly used to train and run deep learning models.
Best for everyday computing.	Best for graphics and high-speed parallel processing.	Best for AI and neural network processing.

Q9. Who won last year's Turing Award?

Answer:
The 2025 A.M. Turing Award was awarded to Charles H. Bennett and Gilles Brassard for their pioneering contributions to quantum information science.

Q10. What is the Stampeding Herd Problem?

Answer:
The Stampeding Herd Problem happens when many users or processes try to access the same resource at the same time. This sudden increase in requests can slow down or overload the system. It can be reduced by using caching, load balancing, and proper request management.

Q11. Who are the founders of Intel and AMD?

Answer:
Intel: Robert Noyce and Gordon Moore
AMD: Jerry Sanders and his team

Q12. Which CPU architecture does your laptop use?

Answer:
My laptop uses a 64-bit (x64-based) CPU architecture with an Intel Core i5-1240H processor. It supports modern operating systems and provides good performance for multitasking.

Q13. What is *The Imitation Game* movie? What did you learn from it?

Answer:
The Imitation Game is a movie based on the life of Alan Turing, a British mathematician and computer scientist. It shows how he and his team worked to break the German Enigma code during World War II. The movie also explains his important contribution to the development of modern computers.

From this movie, I learned that logical thinking, patience, and teamwork can solve difficult problems. I also learned that Alan Turing's work became the foundation of

computer science and artificial intelligence. The movie inspired me to think creatively, work hard, and never give up when facing challenges.